

RENATO SPRENGER-VUKOVIĆ

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SUMMARY

PhD candidate in Computer Science researching large language models, information extraction, and knowledge structuring. Author of multiple peer-reviewed publications at top-tier venues and experienced in teaching and supervising undergraduate and graduate research. Furthermore, I have experience in applying my research to real-world problems through freelance projects in industry, demonstrating the practical impact of my work.

EXPERIENCE

Doctoral Researcher

Heinrich Heine University Düsseldorf

Jan 2023 – Jan 2027

Düsseldorf, Germany

- Conducted doctoral research on large language models and information extraction.
- Supervised Bachelor's and Master's thesis projects.

Freelance Machine Learning Engineer

Deutscher Versand Service

Feb 2024 – Nov 2024

Remote

- Designed and deployed a GDPR-compliant LLM-based customer service system, reducing manual workload.

Freelance Data Analyst

Westdeutscher Rundfunk

Jan 2025 – May 2025

Remote

- Developed a statistical filtering pipeline to detect anomalous streaming activity, reducing daily variance by up to 50%.

SKILLS

Programming

Python (advanced), Java, C++

Machine Learning

PyTorch, HuggingFace, structured prediction, model evaluation, fine-tuning, prompting

Mathematics

Probability, statistics, linear algebra, hypothesis testing, optimization

Software Engineering

Object-oriented design, test-driven development, Git

Cloud

Google Cloud Platform, Microsoft Azure

EDUCATION

PhD in Computer Science, Heinrich Heine University

2023 – 2027 (expected)

Large Language Models, Information Extraction

Master of Computer Science, Heinrich Heine University

2019 – 2022

Bachelor of Computer Science, Minor in Physics, Heinrich Heine University

2015 – 2019

SELECTED PUBLICATIONS

Text-to-SQL Task-oriented Dialogue Ontology Construction — TACL 2026

Post-Training Large Language Models via Reinforcement Learning from Self-Feedback — Preprint arXiv 2025

Dialogue Ontology Relation Extraction via Constrained Chain-of-Thought Decoding — SIGDIAL 2024

Less is More: Local Intrinsic Dimensions of Contextual Language Models — NeurIPS 2025